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REVIEW

The application of low-level laser in the treatment of periodontal Intra-bony defects: systematic review and meta-analysis

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Abstract

This study aimed to evaluate the efficacy of low-level laser application in the treatment of periodontal intra-bony defects. Two independent reviewers conducted searches for articles published in English. The primary objective was to investigate whether clinical parameters improved in periodontal regeneration therapy with and without low-level laser therapy (LLLT). The primary outcome was probing depth (PD), while the secondary outcome measures were clinical attachment level (CAL), gingival recession (GR), and bone defect depth. After screening and eligibility assessment, eight studies were included in the analysis. For the primary outcome measure (PD), after sensitivity analysis, the reduction in PD from baseline to 3 months was greater in the laser group than in the non-laser group ($P=0.02$). For the secondary outcome measure (CAL), after sensitivity analysis, the increase in CAL from baseline to 3 months was greater in the laser group ($P<0.001$). Although laser therapy may be beneficial for early wound healing, the current evidence does not support its routine use in periodontal regenerative surgery. Further high-quality studies are warranted to evaluate its clinical efficacy.

Keywords Low-level laser therapy · Periodontitis · Intra-bony defects · Periodontal regenerative surgery · Meta-analysis

Introduction

Periodontitis is an inflammatory disease that affects the supporting tissues of the teeth. When periodontitis progresses to stages III and IV, it leads to severe resorption of the alveolar bone, resulting in tooth mobility and eventual tooth loss. This significantly impacts the patient's chewing function and quality of life [1]. The primary treatment approach for periodontitis involves controlling inflammation, promoting periodontal tissue regeneration, and ensuring regular maintenance. Among these strategies, periodontal tissue regeneration is considered the most optimal, yet also the most challenging form of treatment [2].

Currently, periodontal tissue regeneration primarily relies on surgical interventions, including guided tissue regeneration (GTR), bone grafting procedures, and the use of bioactive materials to promote regeneration, often in combination [3]. The success of periodontal regenerative surgery is influenced by various factors, with soft tissue healing playing a pivotal role in the overall outcome [4, 5].

In recent years, lasers have increasingly been applied in oral treatments [6]. Low-level laser therapy (LLLT) mainly exerts anti-inflammatory, biostimulatory, and analgesic effects, commonly referred to as photobiomodulation [7]. LLLT has been shown to promote osteoblast proliferation and calcification, modulate osteoblast gene expression, enhance fibroblast proliferation and migration, and inhibit inflammation-related gene expression within cells [8, 9]. Clinical studies have demonstrated that combining scaling and root planing (SRP) with semiconductor laser therapy

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significantly reduces the detection rate of Porphyromonas gingivalis on the inner wall of periodontal pockets [10]. In patients with systemic diseases, LLLT can also reduce bacterial load in periodontal pockets and facilitate the healing of deep periodontal pockets [11]. Furthermore, semiconductor lasers have been found to positively impact the reduction of periodontal pockets, enhance attachment gain, accelerate healing, and alleviate postoperative pain in periodontal flap surgery [12].

LLLT promotes soft tissue healing by stimulating fibroblast and osteoblast proliferation, while also exhibiting antibacterial and analgesic effects. Despite its potential benefits, the null hypothesis posits that LLLT does not result in significant improvements in clinical outcomes when compared to conventional periodontal regenerative surgery.

Materials and methods

The protocol for this systematic review was registered on PROSPERO on March 20, 2025, with the ID: CRD420251012627.

Primary and secondary outcomes

The primary outcome measure of this systematic review was the probing depth (PD) after the regeneration treatment of periodontal intraosseous defects assisted by low-level laser therapy.

The secondary outcome measures were the changes in clinical attachment loss (CAL), gingival recession (GR), and depth of bone defect after low-level laser-assisted regeneration treatment of periodontal intraosseous defects.

Inclusion/exclusion criteria

The PICOS principle was used as the basis for the inclusion criteria.

P: Patients who have learned self-oral hygiene maintenance and still have probing depths of ≥ 5 mm and intra-bony defect depths of ≥3 mm after comprehensive periodontal basic treatment, and meet the indications for periodontal regeneration surgery.

I: Low-energy laser combined with periodontal regeneration surgery.

C: Perform periodontal regeneration surgery alone.

O: Probing depth, Clinical attachment loss, gingival recession, depth of bone defect.

S: Randomized controlled clinical trial.

Exclusion criteria: smokers; patients with systemic diseases such as diabetes, hypertension, coronary heart disease, and atherosclerosis; pregnant and lactating women; articles published in languages other than English.

Search strategy

By April 20, 2025, four electronic databases were searched: PubMed, Cochrane Library, Web of Science, and Embase. The search terms are shown in Table 1. Grey literature sources (e.g., conference abstracts, dissertations) were excluded. Only studies published in English were included.

Study screening

Two reviewers independently screened the titles and abstracts of all included studies. If they could not reach a consensus on any article, they would read the full text for screening. If they still could not determine the article, a third reviewer would be invited to evaluate it. Through the initial assessment of titles and abstracts, irrelevant studies were excluded. After strict screening of the study, the full text was read again for screening. The article screening process is shown in Fig. 1.

Data extraction

For the articles determined to be included, the following information was collected from the studies: statistical research type, sample size, research design, periodontal index, description of bone defect, surgical method, materials used during surgery, method and frequency of laser application, and postoperative care. The data were independently collected by two researchers and recorded in Table 2.

Quality evaluation of included studies

The risk of bias in the included studies was evaluated using version 5.1.0 of the Cochrane Handbook for Systematic Reviews. Seven items were included: random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, other bias. Two independent reviewers conducted the evaluation. The quality assessment of the included articles is detailed in Fig. 2.

Statistical analysis

Meta-analysis of the included study was conducted using RevMan 5.4 software. The primary and secondary

Table 1 Search strategy for each database

Database	Search strategy	Number of results
Pubmed	#1 ((((bone loss, periodontal[MeSH Terms]) OR (periodontitis [Title/Abstract])) OR (Aggressive periodontitis [Title/Abstract])) OR (chronic periodontitis [Title/Abstract])) OR (periodontal disease [Title/Abstract]) #2 (((((periodontal surgery [Title/Abstract]) OR (periodontal operation [Title/Abstract])) OR (periodontal surgical therapy [Title/Abstract])) OR (periodontic surgery [Title/Abstract])) OR (periodontal tissue regeneration [Title/Abstract])) OR (guided tissue regeneration [Title/Abstract]) #3 (((((((((Diode laser[MeSH Terms]) OR (carbon dioxide lasers[MeSH Terms])) OR (erbium YAG laser[MeSH Terms])) OR (ablation, laser tissue[MeSH Terms])) OR (doppler laser flowmetry[MeSH Terms]))) OR (Laser[Title/Abstract])) OR (Low-level laser[Title/Abstract])) OR (Semiconductor[Title/Abstract])) OR (Neodymium YAG laser[Title/Abstract]) #4 #1 AND #2 AND #3	31
Web of science	#1 (((((TS= (bone loss, periodontal)) OR TS=(periodontitis)) OR TS= (Aggressive periodontitis)) OR TS= (chronic periodontitis)) OR TS= (periodontal disease)) #2 (((((TS= (periodontal surgery)) OR TS= (periodontal operation)) OR TS= (periodontal surgical therapy)) OR TS= (periodontic surgery)) OR TS= (periodontal tissue regeneration)) OR TS= (guided tissue regeneration) #3 (((((((TS= (Diode laser)) OR TS= (carbon dioxide lasers)) OR TS= (erbium YAG laser)) OR TS= (ablation, laser tissue)) OR TS= (doppler laser flowmetry)) OR TS=(Laser)) OR TS= (Low-level laser)) OR TS=(Semiconductor)) OR TS= (Neodymium YAG laser) #4 (#1 AND #2 AND #3) AND TS= (randomized controlled trial)	181
Cochrane	#1 (bone loss, periodontal): ti, ab, kw OR (periodontitis): ti, ab, kw OR (Aggressive periodontitis): ti, ab, kw OR (chronic periodontitis): ti, ab, kw OR (periodontal disease): ti, ab, kw #2 (periodontal surgery): ti, ab, kw OR (periodontal operation): ti, ab, kw OR (periodontal surgical therapy): ti, ab, kw OR (periodontic surgery): ti, ab, kw OR (periodontal tissue regeneration): ti, ab, kw OR (guided tissue regeneration): ti, ab, kw #3 (Diode laser): ti, ab, kw OR (carbon dioxide lasers): ti, ab, kw OR (erbium YAG laser): ti, ab, kw OR (ablation, laser tissue): ti, ab, kw OR (doppler laser flowmetry): ti, ab, kw OR (Laser): ti, ab, kw OR (Low-level laser): ti, ab, kw OR (Semiconductor): ti, ab, kw OR (Neodymium YAG laser): ti, ab, kw #4 #1AND #2 AND #3	320
Embase	#1 'bone loss, periodontal' OR 'periodontitis'/exp OR 'periodontitis' OR 'aggressive periodontitis'/exp OR 'aggressive periodontitis' OR 'chronic periodontitis'/exp OR 'chronic periodontitis' OR 'periodontal disease'/exp OR 'periodontal disease' #2 'periodontal surgery' OR 'periodontal operation' OR 'periodontal surgical therapy' OR 'periodontic surgery' OR 'periodontal tissue regeneration' OR 'guided tissue regeneration' #3 'diode laser' OR 'carbon dioxide lasers' OR 'erbium yag laser' OR 'ablation, laser tissue' OR 'doppler laser flowmetry' OR 'laser' OR 'low-level laser' OR 'semiconductor' OR 'neodymium yag laser' #4 #1AND #2 AND #3	94

outcome measures in this study were continuous variables, expressed as mean ± standard deviation. If the statistical result showed $P > 0.05$ and $I^2 \leq 50\%$, it indicated no significant statistical difference or low heterogeneity among the studies, and thus a fixed-effect model was used for data analysis. Conversely, if $P \leq 0.05$ and $I^2 > 50\%$, it indicated significant statistical differences among the studies, and a random-effect model was adopted for analysis.

Results

Study search and screening results

Initially, a total of 626 articles were screened from the database, along with one additional article selected manually [16], resulting in a cumulative total of 627 articles.

After reading the titles and abstracts, 14 potentially relevant articles were identified [14–27]. The full texts of these

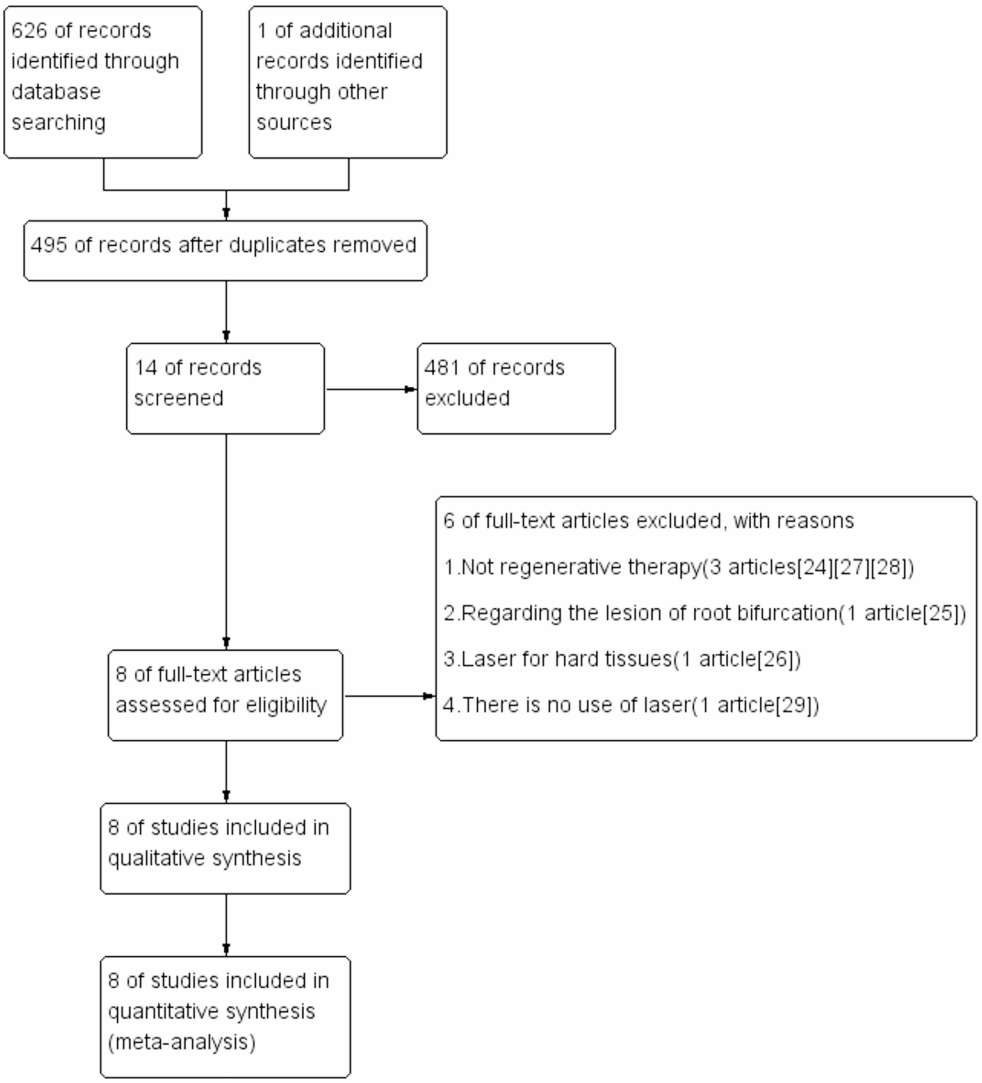


Fig. 1 Flow chart of search

articles were obtained and carefully reviewed and screened. Ultimately, 8 articles were included in the systematic review [14–21]. The other 6 articles [22–27] were excluded, and the reasons for exclusion are shown in Fig. 1.

Characteristics of the included study

A total of 8 articles were included in this study and were classified into three groups based on the materials used for bone defect filling: GTR group [17], Bone grafts group [15, 18, 19], and Other methods group [14, 16, 20, 21]. Among these 8 articles, 7 were randomized controlled trials with split-mouth design and 1 [16] was a randomized controlled trial with parallel design. The types of lasers used were semiconductor lasers or Nd: YAG lasers. The application of lasers was either after surgical debridement or at different times postoperatively. Regarding the surgical methods, 7 articles employed flap surgery, while 1 [16] article utilized the simplified papilla preservation flap technique. All 8 articles provided corresponding postoperative care. For detailed information, please refer to Table 2.

Quality evaluation of included studies

Eight studies used randomization to group the subjects. However, none of the eight studies mentioned the allocation concealment or the blinding of participants and personnel. Regarding the blinding of outcome assessment, two studies [15, 20] did not mention it, while the rest of the studies did involve it.3.4 Meta-analysis results.

Table 2 Basic information of the included study

Treatment and Reference	design of experiment	Group		Patients (*n*)	Age(years)	Number of Defects		Type of Defect	Laser			Surgery Type		Postoperative care	Follow-Up(months)
		Test	Control			Test	Control		Type	Wavelength	The time of laser application	Test	Control		
GTR group															
Doğan et al. 2014[17]	RCT+split-mouth	GTR+L	GTR	13	26–49	13	13	PD≥5 mm, IBD≥3 mm	Nd: YAG	1064 nm,100mW, 100 mJ	At the time of Surgery and on days 1,3,5,7postoperatively	GTR+L	GTR	Amoxicillin 500 mg, three times daily for 1 week,0.2%chlorhexidine mouth rinse	6
Bone grafts group															
Vinaya et al.2023[18]	RCT+split-mouth	HA+β-TCP+L	HA+β-TCP	40	35–60	40	40	PD≥6 mm, IBD	Diode	810 nm,2.5 W	At the time of Surgery	OFD+HA+β-TCP+L	OFD+HA+β-TCP	Amoxicillin 500 mg, three times daily for 5 days, 0.12%chlorhexidine mouth rinse for 2 weeks	6
Gangolu et al.2022[15]	RCT+split-mouth	HA+L	HA	12	20–65	12	12	PD≥5 mm, IBD≥3 mm	Diode	980 nm,3.0 W,2.5 kHz	At the time of Surgery	OFD+HA+L+magnification	OFD+HA+m-agnification	Amoxicillin 500 mg, three times daily+diclofenac 50 mg, twice daily, 0.2%chlorhexidine mouth rinse for 2 weeks	6
AboElsaad N. et al.2009[19]	RCT+split-mouth	Bioactive glass+L	Bioactive glass	20	33–57	20	20	PD≥6 mm, IBD≥3 mm	Diode	830 nm,40mW,16 J/cm^2	At the time of Surgery and on days 3,5,7postoperatively	OFD+Bioactive glass+L	OFD+Bioacti-ve glass	Amoxicillin 500 mg, three times daily for 1 week, 0.2%chlorhexidine mouth rinse for 2 weeks	6
Other methods group															
Dharani et al.2024[20]	RCT+split-mouth	CGF+L	CGF	10	20–50	10	10	PD≥5 mm, IBD	Diode	810±20 nm,1 W,4 J/cm^2	At the time of Surgery	OFD+CGF+L	OFD+CGF	antibiotics and analgesics for 5 days	6
Thalaimalai et al.2020[16]	RCT	PRF+L	PRF	30	ND	15	15	PD≥5 mm, IBD≥3 mm	Diode	810 nm,0.5 W	At the time of Surgery	SPPF+IMP+PRF+L	SPPF+PRF	Amoxycillin 500 mg three times daily, ibuprofen 400 mg three times daily, 0.12%chlorhexidine mouth rinse for 2 weeks	6
Dilsiz et al.2010[21]	RCT+split-mouth	EMP+L	EMP	21	33–46	21	21	PD≥6 mm, IBD≥3 mm	Nd: YAG	1064 nm,1 W,10 Hz,100 mJ	At the time of Surgery	OFD+EMP+L	OFD+EMP+EDTA	Naproxen,550 mg, twice daily for 5 days,0.12%chlorhexidine mouth rinse for 2 weeks	12
Ozcelik et al.2008[14]	RCT+split-mouth	EMD+L	EMD	22	31–49	22	22	PD≥6 mm, IBD≥3 mm	Diode	588 nm, 4 J/cm^2	At the time of Surgery and on days 1,2,3,4,5postoperatively	OFD+EMD+L	OFD+EMD	Doxycycline 100 mg twice daily for 1 week, 0.12%chlorhexidine mouth rinse for 2 weeks	12

RCT: randomized controlled trial; GTR: guided tissue regeneration; L:Laser; PD: probing depth; IBD: infra-bony defect; HA: hydroxyapatite; β-TCP: β-tricalcium phosphate; OFD: open flap debridement; CGF: concentrated growth factor; PRF: plate-rich fibrin; SPPF: simplified papilla preservation flap; IMP: intra-marrow penetration; EMP: enamel matrix proteins; EMD: enamel matrix protein derivative; ND: not defined

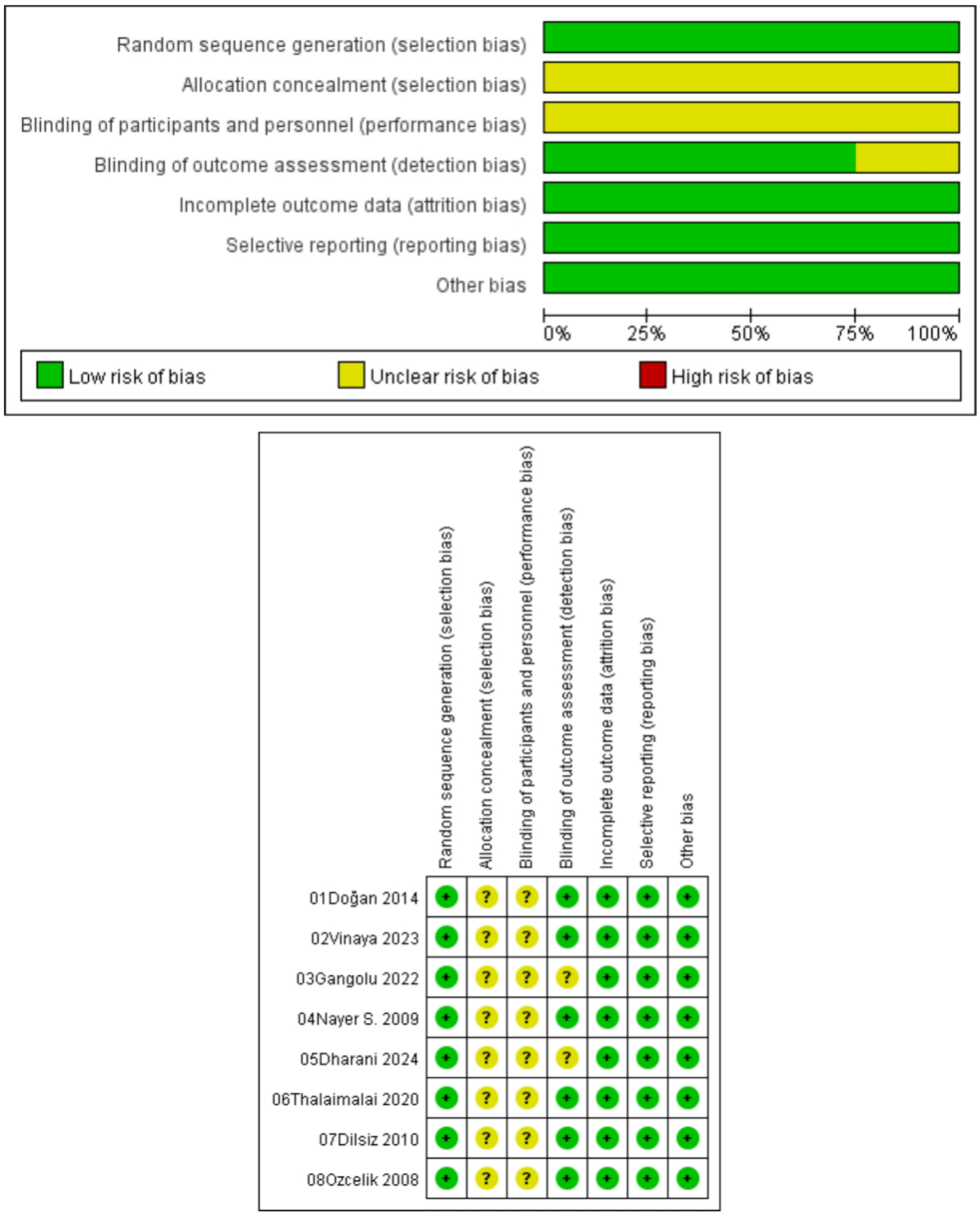


Fig. 2 Risk bias graph of the included study

Meta-analysis results

PD

Among the 8 included articles, 5 [15, 17–20] analyzed the changes in PD from baseline to 3 months. The mean difference (MD) value of PD in periodontal regeneration treatment with and without laser irradiation was 0.34 mm (95% CI = −0.08 to 0.76 mm, $P = 0.11$), as shown in Fig. 3A. There was no statistically significant difference between the two groups.

Among the 8 included articles, 7 [15–21] analyzed the changes in PD from baseline to 6 months (Table 3). In periodontal regeneration treatment with and without laser irradiation, the MD value of PD was 0.08 mm (95% CI = −0.56 to 0.73 mm, $P = 0.80$), as shown in Fig. 3B. There was no statistically significant difference between the two groups. However, in the subgroup analysis, the MD value of the Bone grafts group [15, 18, 19] was − 0.17 mm (95% CI = −0.25 to −0.09 mm, $P < 0.0001$). It refers to the mean PD reducation in the laser group was 0.17 mm lower than that in the non-laser group at the 6-month follow-up, and the difference was statistically significant.

CAL

Among the 8 included articles, 5 [15, 17–20] analyzed the changes in CAL gain from baseline to 3 months (Table 3).

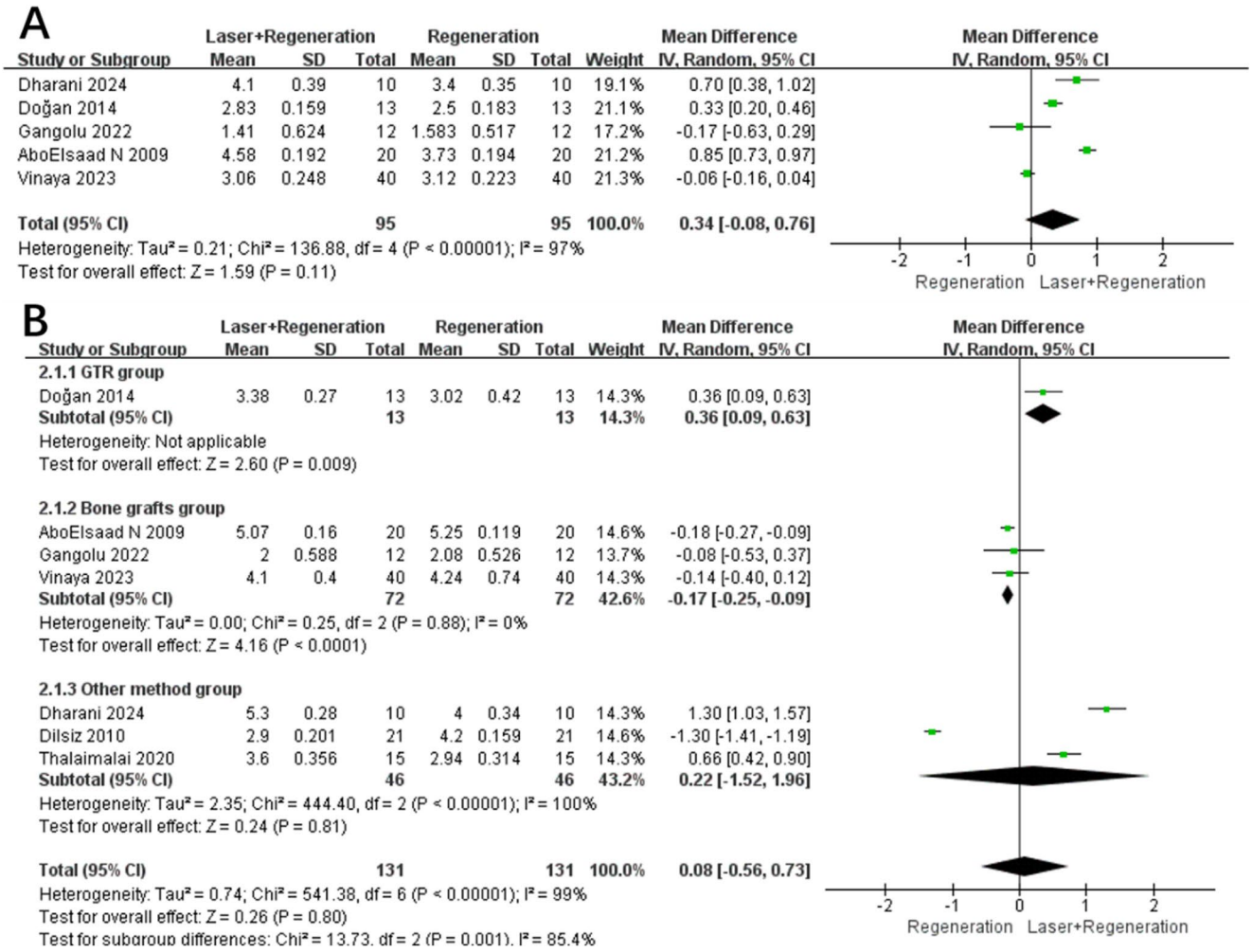


Fig. 3 (**A**) For periodontal regeneration surgeries with or without laser treatment, a meta-analysis was conducted for the comparison of PD at baseline and 3 months. The overall MD was 0.34 mm (95%CI=−0.08 to 0.76 mm, P=0.11). **B** For periodontal regeneration surgeries with or without laser treatment, a meta-analysis was conducted for the comparison of PD at baseline and 6 months. The overall MD was 0.08 mm (95%CI=−0.56 to 0.73 mm, P=0.80)

In the laser group, the average CAL gain was 0.66 mm (95%CI=−0.03 to 1.35 mm, P = 0.06) more than that in the non-laser group, and the difference was not statistically significant.

Among the 8 included articles, 7 [15–21] analyzed the changes in CAL from baseline to 6 months. The MD value of CAL in laser group and non-laser group was 0.41 mm (95% CI = −1.04 to 1.86 mm, P = 0.58), as shown in Fig. 4B. There was no statistically significant difference between the two groups.

GR

Among the 8 included articles, 4 [16–18, 21] analyzed the changes in GR from baseline to 6 months (Table 4). In periodontal regeneration therapy with and without laser irradiation, the MD value of GR was − 0.04 mm (95% CI = −0.33 to 0.25 mm, P = 0.79), as shown in Fig. 5. There was no statistically significant difference between the two groups.

Intra-bony defects depth

Among the 8 included articles, 3 [15, 16, 19] analyzed the changes in the depth of intra-bony defects from baseline to 6 months (Table 4). In periodontal regenerative therapy with and without laser irradiation, the MD value of intra-bony defects depth was 0.98 mm (95% CI = −0.09 to 2.05 mm, P = 0.07), as shown in Fig. 6. P >0.05, indicating no statistically significant difference between the two groups.

Sensitivity analysis

The sensitivity was verified by the method of eliminating study one by one. It was found that in the analysis of the

Table 3 Quantitative assessment for regenerative surgery versus laser: PD and CAL

Treatment and Reference	Mean PD (mm)								Mean CAL (mm)							
	Baseline		3 month		6 month		12month	Reduction	Baseline		3 month		6 month		12 month	Reduction
	Test	Control	Test	Control	Test	Control	Test Control	Test Control	Test	Control	Test	Control	Test	Control	Test Control	Test Control
GTR group																
Doğan et al.2014[17]	6.01±0.47	5.95±0.43	3.18±0.33	3.45±0.50	2.63±0.29	2.93±1.98	ND ND	3.38±0.27 3.02±0.42	7.35±0.48	7.43±0.60	4.98±0.54	5.43±0.76	4.46±0.57	5.18±0.46	ND ND	2.91±0.43 2.26±0.34
Bone grafts group																
Vinaya et al.2023[18]	6.62±1.30	6.76±1.26	3.56±0.88	3.64±0.64	2.52±0.38	2.52±0.42	ND ND	4.10±0.40 4.24±0.74	9.85±1.58	10.02±1.94	7.50±1.90	9.51±2.24	6.64±1.60	8.78±1.98	ND ND	3.21±0.02 1.24±0.04
Gangolu et al.2022[15]	6.08±1.44	6.75±1.2	4.67±1.61	5.167±1.33	4.08±1.44	4.67±1.37	ND ND	ND ND	6.583±2.39	7.75±2.17	5.16±2.51	6.167±2.32	4.58±2.39	5.67±2.42	ND ND	ND ND
AboElsaad N.et al.2009[19]	8.48±0.5	8.83±0.34	3.9±0.7	5.1±0.8	3.41±0.51	3.58±0.41	ND ND	ND ND	7.21±0.35	7.75±0.96	4.85±0.65	5.78±0.88	5±0.85	5.15±0.53	ND ND	ND ND
Other methods group																
Dharani et al.2024[20]	7.30±0.949	7.30±0.823	3.20±0.789	3.90±0.738	2.00±0.667	3.30±0.483	ND ND	5.3±0.28 4.0±0.34	7.20±1.549	6.80±1.033	3.10±1.287	3.40±1.075	1.90±1.101	2.80±0.789	ND ND	5.30±0.44 4.0±0.25
Thalaimalai et al.2020[16]	7.53±1.125	7.07±1.033	ND	ND	3.93±0.799	4.13±0.640	ND ND	ND ND	9.07±1.438	8.73±1.387	ND	ND	5.8±1.568	6.47±1.362	ND ND	ND ND
Dilsiz et al.2010[21]	7.3±0.6	7.3±0.7	ND	ND	4.4±0.7	3.1±0.2	3.3±0.4 3±0.4	4±0.8 4.4±0.9	9.5±0.7	9.3±0.8	ND	ND	7.9±0.8	6.1±0.4	6.9±0.7 6.4±0.5	2.6±1.2 3±1.1
Ozcelik et al.2008[14]	6.6(5.9–6.8)	6.5(5.9–6.7)	ND	ND	2.5(2.2–3.9)	2.9(2.3–3.7)	1.3(1.2–2.3) 1.7(1.3–3.3)	ND ND	7.2(6.8–7.5)	7.2(6.5–7.5)	ND	ND	4.4(3.0–6.0)	4.4(3.5–6.5)	3.1(2.4–3.7) 3.2(2.8–4.7)	ND ND

ND: not defined

change from baseline to 3 months in PD, after removing the article Vinaya et al. 2023 [18], the P value changed from 0.11 to 0.02, indicating a statistically significant difference. In the analysis of the change from baseline to 3 months in CAL, after still removing the article Vinaya et al. 2023 [18], I^2 decreased from 98% to 15%, and $P < 0.001$, indicating a statistically significant difference. In the analysis of the change from baseline to 6 months in the depth of bone defect, after removing the article AboElsaad N. et al. 2009 [19], I^2 decreased from 98% to 71%, and $P < 0.001$ (Table 5).

Discussion

Low-level laser therapy (LLLT) has been shown to promote tissue cell proliferation, inhibit inflammation, and reduce bone resorption [8]. In recent years, its use has been increasingly adopted by dental professionals. In the adjuvant treatment of periodontal flap surgery, studies have found that LLLT significantly reduces probing depth (PD), increases clinical attachment level (CAL), and alleviates postoperative pain [12]. In this study, however, no significant differences were observed in PD and CAL values at 3 and 6 months after periodontal regenerative surgery between the LLLT group and the control group.

Sensitivity analysis revealed that, after excluding the study by Vinaya et al. (2023) [18], the reduction in PD and the increase in CAL from baseline to 3 months in the LLLT group were more favorable compared to the control group, showing statistical significance and reduced heterogeneity. In the analysis of gingival recession (GR), no additional benefit was found for the laser treatment. Regarding the depth of intra-bony defects, the average bone fill in the LLLT group was 0.98 mm [95% CI = −0.09, 2.05 mm, P = 0.07] greater than in the non-laser group after periodontal regenerative surgery, though this difference was not statistically significant.

Summary of PD reduction

The MD from baseline to 3 months was 0.34 mm [95% CI = −0.08, 0.76 mm, P = 0.11], and, when compared to the MD value from baseline to 6 months of 0.08 mm [95% CI = −0.56, 0.73 mm, P = 0.8], it was found that the use of low-level laser therapy (LLLT) resulted in a 0.34 mm greater reduction in periodontal pocket depth at 3 months after periodontal regenerative surgery compared to those who underwent the surgery alone. This result is consistent with studies investigating LLLT intervention in periodontal flap surgery, indicating that LLLT may aid in early wound healing [28, 29]. However, its effectiveness appears limited,

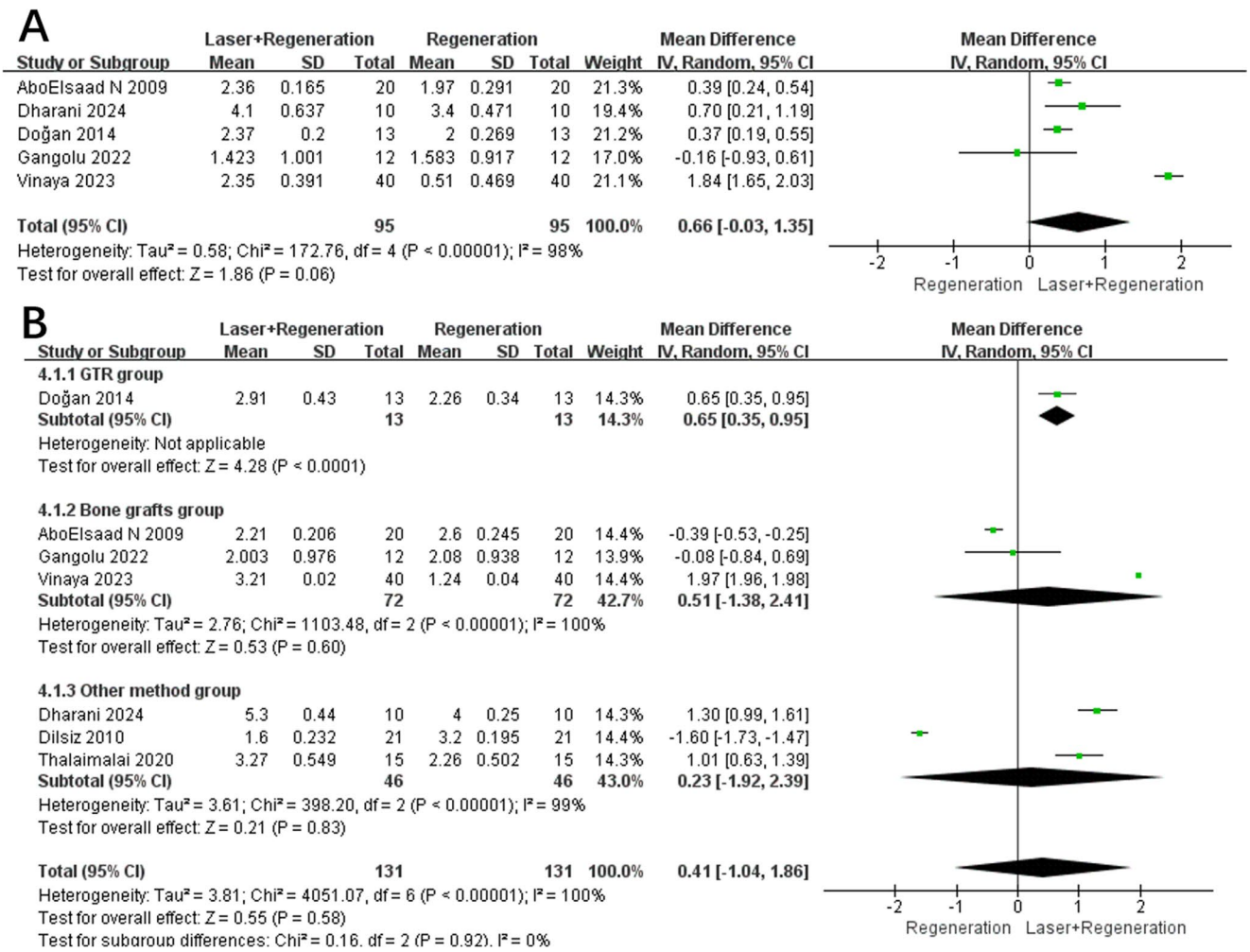


Fig. 4 (**A**) For periodontal regeneration surgeries with or without laser treatment, a meta-analysis was conducted for the comparison of CAL at baseline and 3 months. The overall MD was 0.66 mm (95%CI=−0.03 to 1.35 mm, $P=0.06$). (**B**) For periodontal regeneration surgeries with or without laser treatment, a meta-analysis was conducted for the comparison of CAL at baseline and 6 months. The overall MD was 0.41 mm (95%CI=−1.04 to 1.86 mm, $P=0.58$)

with a change in probing depth (PD) of less than 0.5 mm, resulting in no significant clinical effect. After 6 months or longer post-surgery, the depth of periodontal pockets and attachment levels may be influenced by patient factors such as oral hygiene compliance, systemic disease status, susceptibility to disease progression, and participation in professional periodontal care programs [5, 30–32].

The gingival tissues, particularly the junctional epithelium and the underlying connective tissue, serve as a soft tissue seal for the deeper periodontal structures. Therefore, promoting soft tissue healing in the early postoperative period can facilitate the osteogenesis process of bone grafts or bioactive materials placed in the apical region. In vitro studies have demonstrated that laser irradiation with energy densities ranging from 1 to 10 J/cm² significantly enhances cellular proliferation [8].In studies involving primary human gingival fibroblasts, semiconductor lasers have been shown to significantly improve cell viability and enhance migratory capacity [33]. Furthermore, in experiments using human gingival fibroblasts stimulated by lipopolysaccharide (LPS), semiconductor laser irradiation was found to suppress LPS-induced prostaglandin E2 (PGE2) production by downregulating cyclooxygenase-2 (COX-2) mRNA expression, thereby reducing the inflammatory response [34].

Summary of CAL gain

No statistically significant differences were observed in the changes from baseline to 3 months and 6 months. In the

Table 4 Quantitative assessment for regenerative surgery versus laser: gingival recession and Intra-bony defects depth

Treatment and Reference	Gingival recession (mm)								Intra-bony defects depth(mm)							
	Baseline		3 month		6 month		12month	Change	Baseline		3 month		6 month		12 month	
	Test	Control	Test	Control	Test	Control	Test Control	Test Control	Test	Control	Test	Control	Test	Control	Test	Control
GTR group																
Doğan et al. 2014[17]	1.33±0.42	1.48±0.55	1.80±0.42	1.98±0.46	1.82±0.52	2.24±0.43	ND ND	ND ND	ND	ND	ND	ND	ND	ND	ND	ND
Bone grafts group																
Vinaya et al. 2023[18]	4.34±0.96	4.42±1.18	4.02±1.36	4.18±1.24	4.04±1.28	4.20±1.36	ND ND	1.30±0.22 0.20±0.38	ND	ND	ND	ND	ND	ND	ND	ND
Gangolu et al. 2022[15]	ND	ND	ND	ND	ND	ND	ND ND	ND ND	4.84±1.168	4.66±1.28	ND	ND	3.64±1.28	4.16±1.12	ND	ND
AboElsaad N. et al. 2009[19]	ND	ND	ND	ND	ND	ND	ND ND	ND ND	5.1±0.5	5.1±0.5	2.2 ±0.6	3.2±0.4	2.1±0.4	2.3±0.5	ND	ND
Other methods group																
Dharani et al. 2024[20]	ND	ND	ND	ND	ND	ND	ND ND	ND ND	ND	ND	ND	ND	ND	ND	ND	ND
Thalaimalai et al. 2020[16]	1.53±0.915	1.67±0.900	ND	ND	1.87±1.060	2.33±1.175	ND ND	ND ND	5.93±1.752	4.42±1.675	ND	ND	3.487±1.008	3.160±1.135	ND	ND
Dilsiz et al. 2010[21]	2.2±0.5	2±0.4	ND	ND	3.5±0.5	3±0.2	3.6±0.5	3.4±0.3 1.5±0.7	1.4±0.6	10.8±0.5	10.8±0.4	ND	ND	ND	ND	ND
Özçelik et al. 2008[14]	1.9(1.1–2.4)	2.3(1.8–3.0.8.0)	ND	ND	2.2(0.9–2.6)	2.6(2.3–3.4)	2.3(1.1–2.8)	2.9(2.4–3.8) ND	ND	ND	ND	ND	ND	ND	ND	ND

ND: not defined

sensitivity analysis, after excluding the study by Vinaya et al. (2023) [18], the difference in clinical attachment level (CAL) became statistically significant, and heterogeneity was reduced. This suggests that, after controlling for inter-study heterogeneity, low-level laser therapy (LLLT) resulted in an additional CAL gain of 0.39 mm at 3 months. However, such a gain is generally considered clinically negligible, indicating only a minor benefit of laser application in this specific clinical context.

The wound surface in periodontal surgery has certain particularities. It does not primarily involve the healing of two vascularized gingival flaps, but rather the attachment of a single gingival flap to the avascular and acellular root surface [35]. Basic research has shown that periodontal wounds typically achieve functional integrity within two weeks after surgery. Therefore, the stability of the wound surface during the early healing stage depends on the suturing effect, but it is crucial to ensure that it is not compromised by plaque re-colonization [35]. Repeated laser irradiation of the wound surface within one week after surgery is beneficial for the reattachment of periodontal tissues and promotes an increase in clinical attachment level (CAL) gain [14, 17, 19]. This effect is similar to the use of chlorhexidine mouthwash after surgery to control oral bacteria and assist in the early stages of wound healing [36].

Summary of GR and intra-bony defects depth

The MD value of GR was − 0.04 mm [95% CI = −0.33, 0.25 mm, $P = 0.79$], indicating that low-level laser therapy did not show any advantage in controlling gingival recession. This result is similar to the previous statistical studies [37].

In articles with detailed descriptions of the depth of bone defects [15, 16, 19], statistical analysis revealed that from baseline to 6 months, the MD value of intra-bony defect depth was 0.98 mm [95% CI = −0.09, 2.05 mm, $P = 0.07$], the difference was not statistically significant. After excluding the article by AboElsaad N. et al. 2009 [19], this advantage became more pronounced at 1.44 mm [95% CI = 0.93, 1.94 mm, $P < 0.001$], and heterogeneity decreased. However, the reliability of this result is limited by the reduced number of articles.

Summary of periodontal regeneration surgery

The undifferentiated mesenchymal cells within the periodontal ligament can induce periodontal regeneration. However, their regenerative capacity is often slower than the rate of inflammatory destruction [38]. Therefore, clinicians employ barrier membranes, bone graft materials, or

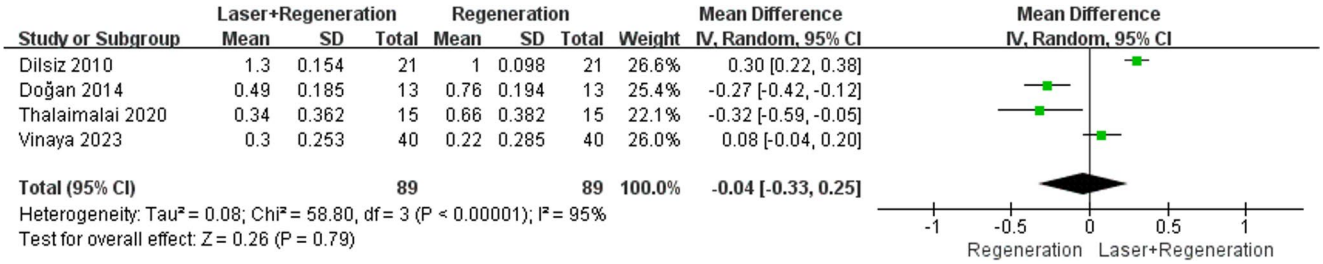


Fig. 5 For periodontal regeneration surgeries with or without laser treatment, a meta-analysis was conducted for the comparison of GR at baseline and 6 months. The overall MD was −0.04 mm (95%CI=−0.33 to 0.25 mm, $P=0.79$)

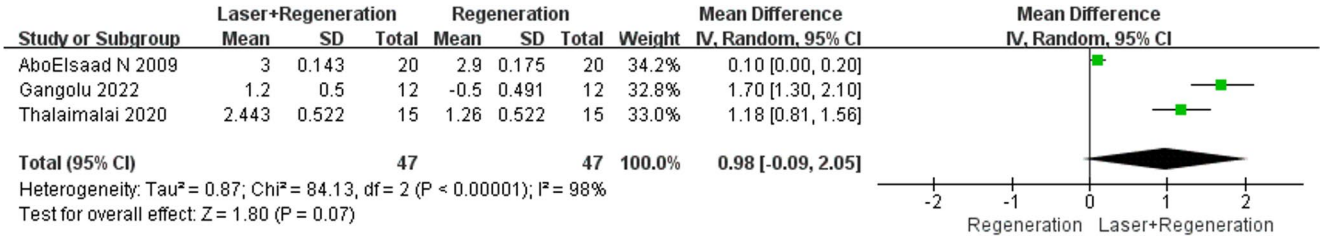


Fig. 6 For periodontal regeneration surgeries with or without laser treatment, a meta-analysis was conducted for the comparison of Intra-bony defects depth at baseline and 6 months. The overall MD was 0.98 mm (95%CI=−0.09 to 2.05 mm, $P=0.07$)

various growth factors to enhance the degree of periodontal tissue regeneration [3, 5, 30–32]. LLLT can increase the content of adenosine triphosphate (ATP) in tissues, promote angiogenesis and collagen fiber synthesis, and alter cell activity by influencing the mitochondrial respiratory chain or membrane calcium channels [39, 40]. In studies on periodontal tissues, it was found that the biological stimulation caused by laser can induce intracellular metabolic changes, accelerate cell division, fibroblast migration and matrix production, and also has an effect on the repair and regeneration of bone metabolism [14, 19, 41]. Research on osteoblasts has shown that low-level lasers can stimulate osteoblast differentiation by regulating cellular responses and increasing the concentrations of alkaline phosphatase, osteocalcin, bone sialoprotein and bone morphogenetic protein, thereby promoting bone formation [42]. Moreover, in newly formed alveolar bone, there was no inflammatory cell infiltration in the group exposed to low-level laser irradiation [43].

Considering the mode of action of laser therapy, clinical applications primarily involve surface irradiation, making it difficult to assess the direct laser stimulation of the bone surface. Therefore, photobiomodulation may primarily support early wound healing rather than bone regeneration.

Limitations of the study

A language restriction was applied, limiting inclusion to English-language studies, which may have affected the comprehensiveness and representativeness of the findings. The exclusion of grey literature may also have introduced publication bias.

Clinical implications

Future investigations should aim to standardize laser parameters and adopt more stringent inclusion criteria to enable a more objective and precise evaluation of laser efficacy in periodontal regeneration surgery.

Conclusions

Current evidence does not support the routine use of LLLT in periodontal regenerative surgery. While LLLT shows some benefit in the early stages of healing, its clinical significance remains minimal. Further high-quality randomized controlled trials (RCTs) are needed to standardize laser parameters and assess long-term outcomes.

Table 5 Details of sensitivity analysis

Comparison	Outcome			Remove study	New outcome		
	Mean difference	I^2	P		Mean difference	I^2	P
PD baseline to 3 months	0.34[−0.08,0.76]	97%	0.11	Doğan et al. 2014[17]	0.34[−0.26,0.93]	98%	0.26
				Vinaya et al. 2023[18]	0.46[0.08,0.84]	93%	0.02*
				Gangolu et al. 2022[15]	0.45[−0.02,0.91]	98%	0.06
				AboElsaad N. et al. 2009[19]	0.21[−0.12,0.53]	91%	0.21
				Dharani et al. 2024[20]	0.26[−0.22,0.73]	98%	0.29
PD baseline to 6 months	0.08[−0.56,0.73]	99%	0.8	Doğan et al. 2014[17]	0.04[−0.67,0.75]	99%	0.91
				Vinaya et al. 2023[18]	0.12[−0.61,0.85]	99%	0.74
				Gangolu et al. 2022[15]	0.11[−0.59,0.82]	99%	0.76
				AboElsaad N. et al. 2009[19]	0.13[−0.82,1.08]	99%	0.79
				Dharani et al. 2024[20]	−0.12[−0.74,0.50]	99%	0.71
				Thalaimalai et al. 2020[16]	−0.01[−0.7,0.68]	99%	0.97
				Dilsiz et al. 2010[21]	0.32[−0.16,0.80]	96%	0.19
AL baseline to 3 months	0.66[−0.33,1.35]	98%	0.06	Doğan et al. 2014[17]	0.72[−0.22,1.67]	98%	0.13
				Vinaya et al. 2023[18]	0.39[0.26,0.52]	15%*	<0.001*
				Gangolu et al. 2022[15]	0.83[0.06,1.59]	98%	0.03*
				AboElsaad N. et al. 2009[19]	0.72[−0.24,1.68]	98%	0.14
				Dharani et al. 2024[20]	0.65[−0.16,1.45]	98%	0.11
AL baseline to 6 months	0.41[−1.04,1.86]	100%	0.58	Doğan et al. 2014[17]	0.37[−1.25,1.99]	100%	0.65
				Vinaya et al. 2023[18]	0.15[−0.82,1.11]	99%	0.77
				Gangolu et al. 2022[15]	0.49[−1.08,2.06]	100%	0.54
				AboElsaad N. et al. 2009[19]	0.54[−1.15,2.24]	100%	0.53
				Dharani et al. 2024[20]	0.26[−1.36,1.89]	100%	0.75
				Thalaimalai et al. 2020[16]	0.31[−1.30,1.92]	100%	0.7
				Dilsiz et al. 2010[21]	0.75[−0.38,1.89]	100%	0.19
GR baseline to 6 months	−0.04[−0.33,0.25]	95%	0.79	Doğan et al. 2014[17]	0.05[−0.22,0.32]	92%	0.71
				Vinaya et al. 2023[18]	−0.09[−0.54,0.37]	97%	0.71
				Thalaimalai et al. 2020[16]	0.04[−0.27,0.35]	96%	0.80
				Dilsiz et al. 2010[21]	−0.16[−0.43,0.12]	88%	0.26
Intra-bony defects depth at baseline to 6 months	0.98[−0.09,2.05]	98%	0.07	Gangolu et al. 2022[15]	0.63[−0.43,1.69]	97%	0.25
				AboElsaad N. et al. 2009[19]	1.44[0.93,1.94]	71%	<0.001*
				Thalaimalai et al. 2020[16]	0.89[−0.68,2.46]	98%	0.27

* The data has changed significantly

Author contributions Yiru Wei designed the study, conducted the systematic literature search, screened the articles for inclusion, evaluated the quality of articles, developed the methodology, and led the overall coordination of the meta-analysis. She also drafted the initial manuscript.Ruonan Xu conducted this systematic literature search, screened the articles for inclusion, and extracted the data relevant to the analysis.Aishan Yilihamu was responsible for the quality assessment of the included studies and assisted in the drafting of the methodology section.Yuting Chen performed the statistical analysis, interpreted the data, and review the statistical presentation throughout the article. Gulinuer Awuti provided expertise on the clinical implications of the findings and reviewed the final manuscript for important intellectual content.All authors read and approved the final manuscript, and they agree to be accountable for all aspects of the work.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Clinical trial number Not applicable.

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